

STUDY AND EVALUATION SCHEME

Of

Diploma (Engineering)- 3Years

Diploma (Engineering)- Civil Engineering II Year

Semester -III



**BHABHA UNIVERSITY,
BHOPAL (M.P.)**



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Programme – Three Years Diploma in CIVIL ENGG

III Semester

Branch – Civil Engineering

Course Code	Subject	Theory Component								Practical Component						Total Credit	Grand Total of Marks
		LECTURES	CONTINUOUS EVALUATION		END OF THE TERM/ SEMESTER EVALUATION			THEORY CREDIT	PRACTICAL Hrs. Per Week	CONTINUOUS EVALUATION	END OF THE TERM/ SEMESTER EVALUATION						
Hrs. Per Week	TERM WORK, QUIZ, ASSIGNMENT	MID TERM TEST (TWO)		THEORY PAPER			LAB WORK, QUIZ, ASSIGNMENT	PRACTIAL / ORAL EXAMINATION (VIVA)									
		I	II	NO.	MARKS	DURATI ON (Hr.)		NO.	MARKS	DURATI ON (Hr.)							
CE 3001	Surveying	03	10	10	10	01	70	3 hrs	04	06	20	1	30	3 hrs	2	06	150
CE 3002	Material Technology	03	10	10	10	01	70	3 hrs	04	02	20	1	30	3 hrs	2	06	150
CE 3003	Building Construction	04	10	10	10	01	70	3 hrs	04	02	20	1	30	3 hrs	2	06	150
CE 3004	Hydraulics	04	10	10	10	01	70	3 hrs	04	02	20	1	30	3 hrs	2	06	150
CE 3005	Building Drawing	02	10	10	10	01	70	3 hrs	04	06	20	1	30	3 hrs	2	06	150
	TOTAL	16	50	50	50	05	350		20	18	100	05	150		10	30	750

Theory Credits : 20

Theory Mark : 350

Minimum Pass Grade in Theory & Practical 'D'

Practical Credits : 10

Practical Marks : 150

Total Credits : 30

Quiz, Mid Term, Lab. Work : 250

BHABHA UNIVERSITY, BHOPAL

Subject –SURVEYING

Sub Code – CE3001

UNIT-I

TYPES OF SURVEY

Definition. Objects of Surveying. Principles of Surveying. Uses of survey, Classification of Surveying. Primary – Plain, Geodetic. Secondary – Based on Instruments, method, object, Nature of field.

CHAIN & CROSS STAFF SURVEY

Principle of Chain Survey. Study and use of Instruments for linear measurements – chain, Tape, Ranging Rod, arrows, pegs, cross Staff, optical Square, line Ranger. Ranging –Direct and Indirect Ranging Chaining – Plain and sloping grounds. Chain Triangulation – Survey Station and their Selections, factors affecting selection of survey station. Survey lines, Check lines, Tie lines, base line. Taking offsets. long and short offset, degree of offset. Obstacles in chaining. Chain & cross staff Survey for finding area of a field (Numerical problems) Errors in chain Surveying & applying Corrections for chain & Tape (Numerical problems).Conventional signs related to survey.

UNIT-II

COMPASS SURVEY

Principle of Compass Survey. Bearing of lines – Meridian –True, Magnetic, and Arbitrary. Bearing –fore bearing, Back bearing, Whole circle bearing, Quadrennial

bearing system and Reduced bearing, Conversion of bearings, finding included angles from bearings. Prismatic Compass – Component, construction and use. Local attraction, Causes, precautions to be taken to avoid and correction of bearings affected due to local attraction, calculation of included angles. Traversing – traversing by chain and compass. open traverse, closed traverse, check on open and closed traverse. Graphical adjustment for closing error. Numerical problems on calculation of bearings, Angles and local attraction.

UNIT-III

LEVELLING

Definitions, meaning of various terms used in leveling – Level surface, Level line, horizontal line, Vertical line, Datum surface, Reduced level, Bench mark and its types. Dumpy level – Components, Construction, Line of sight, Line of Collimation, Bubble tube axis, leveling Staff – Telescopic and folding type. Foresight, back sight, Intermediate sight, Change point, Height of collimation. Fundamental axes and their relationship. Recording in level book. Temporary adjustments of dumpy level.

Method of Reduction of levels – Height of instrument method and Rise and fall method. Arithmetical checks, Numerical problems, Computation of missing readings. Classifications of leveling - simple, differential, profile, cross sectional, fly and check leveling. Study and use of tilting level & Auto level. Sources and errors in leveling, precautions and difficulties faced in leveling.

UNIT-4

CONTOURING

Definitions – Contour, contour interval, Horizontal equivalent. Characteristics of contours. Method of locating contours. Interpolation of contours. Establishing grade contours. Uses of Contour Maps. Calculation of reservoir capacity by contour map by trapezoidal and prismoidal formula. Interpretation of Typical Contour Sheets.

UNIT-V

AREA AND VOLUME MEASUREMENTS

Construction and use of polar planimeter for measurement of area and simple numerical problems. Study and use of Digital Planimete .Concept of computation of Volume by Trapezoidal and Prismoidal formulae.(No numerical problems)

Reference Books : -

LIST OF REFERENCE BOOKS

S. No.	Title	Author	Publisher
1	Surveying and levelling	N.N.Basak	Tata Mc Graw-Hill
2	Surveying And Levelling part I & II	T .P. Kanetkar & S. V.Kulkarni,	Pune Vidhyarthi Griha Prakashan
3	Surveying And Levelling Vol. I & II	Dr. B. C.Punmiya	Laxmi publication
4	Text Book Of Surveying	S.K.Husain & M.S. Nagaraj,	S. Chand And Company

LIST OF EXPERIMENT (EXPANDABLE):

1. Chain & Compass Traverse Survey – A simple closed traverse of 5-6 sides enclosing a building. Calculation of included angles, locating details and plotting them on A 1 size imperial drawing sheet.

2.Block Contouring – A block of 100 x 150m with spot levels at 10x10m plotting the contours on A-1 size imperial drawing sheet with a contour interval Of 1m.

3.Profile levelling survey – Running a longitudinal section for a length of 500 m for a road/canal /railway alignment. Cross section shall be taken suitably. Plotting plan, L-section and cross section on A1 size imperial drawing sheet.

Note -

1. Group size for Survey Practical work should be maximum 8 Students.
2. Each student from a group should handle the instrument independently to understand the function of different Components and use of the instrument.
3. Drawing, plotting should be considered as part of practical.
4. One full day per project is required for carrying out project work.

BHABHA UNIVERSITY, BHOPAL

Subject –MATERIAL TECHNOLOGY

Sub Code – CE3002

UNIT-I

INTRODUCTION :

Importance of material Technology for Civil Engineer, Engineering materials used in construction.

MASONARY MATERIALS

Building stones- classification of rocks, requirement of good building stone, dressing of stones, quarrying of stones, artificial or cast stones. Bricks – properties of good building bricks, method of preparation of bricks, strength of bricks, proportions of burnt clay bricks , testing of bricks, fly ash bricks

UNIT-II

BINDING MATERIALS

Murum, Properties of Murum for Road work. Lime - Types and properties of lime : Fat lime, Hydraulic Lime, Quick lime. Cement - Different ingredients used for manufacturing cement with their percentage. Physical properties of ordinary Portland cement (OPC), hydration of cement. Physical properties , field test of cement, storing cement at site, effect of storage of cement on properties of cement, Types of cement and their functional uses.

UNIT-III

AGGREGATES

Properties of fine aggregates - Concept of size, shape, surface texture, strength, specific gravity, bulk density, water absorption, surface moisture, soundness, bulking impurities. Determination of fineness modulus & grading zone of sand by sieve analysis, determination of silt content in sand & their specification as per IS 383, . Properties of coarse aggregates - Concept of size, shape, surface texture, water absorption, soundness, specific gravity & bulk density, Determination of fineness modulus. Determination of crushing value, impact value & abrasion value of coarse aggregate, flakiness index & elongation index of coarse aggregate and their specification

MORTARS

Classifications, Functions of mortar, proportions, properties and tests for mortar

CONCRETE

Introduction to concrete - Definition of concrete, necessity of supervision for concreting operation, different grades of concrete (as per provisions of IS 456-2000), minimum grade of concrete for different exposure conditions, minimum grade of concrete for R.C.C. Water cement (w/c) ratio, Definition of w/c ratio, significance of w/c ratio, maximum w/c ratio for different grades of concrete for different exposure conditions. Properties of fresh concrete, Definition of workability, factors affecting workability of concrete.

Determination of workability of concrete by slump cone test, compaction factor test, vee bee consistometer. Range values of workability requirement for different types of concrete works, cohesiveness, segregation, bleeding, creep of concrete. Curing of concrete. Testing of concrete for strength and workability. Properties of hardened concrete

UNIT-IV

TIMBER Difference between wood and timber. Timber based material: use of timber, characteristics of good timber, defects in timber, plywood, particle board, veneer, sun mica, fire mica, nuwood, artificial timber, rubber wood

PAINTS, VARNISHES & COLOURS

Different in gradients used in manufacturing/ preparation of paints, Primers, their different types for steel and timber. Use of paint as protecting surface device for steel surface type of paint used and for wood surface types of paint used.

VARNISH: Method of preparation of varnish, component materials used in varnish.

COLOURS: For decorative purpose and finished purpose use of colour as water base, colour as oil base, Distempers and cement paints.

UNIT-V

STEEL AND ALUMINIUM PRODUCTS

Steel used as Engineering Material in different shapes. Like T- section , Angle section, Channel Section, I-Section steel sheets used in manufacturing of Doors. Aluminum : Used as construction materials.

MISCELLANEOUS

Give the concepts about the other materials which can be used as Engineering Materials like Glass, Rubber, Tar, Emulsion, Bitumen, Glass wool, Use of J bolts, U hooks, Stoneware pipes, Galvanized iron pipes.

Miscellaneous materials

Glass, plastic- P.V.C. pipes used as materials in pipe laying for water supply

purposes, Irrigation etc. Water tanks. fibers, aluminum, steel , galvanized iron, asphalt bitumen etc. micro silica, PVC, CPVC, PPF. Waterproofing and termite proofing materials, admixtures in concrete, bonding agents, epoxy resins, Polishing materials etc. Readymade concrete cover. Readymade ornamental material (wall papers, carpets, radium prints, blocks etc.)

REFERENCE BOOKS:-

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Engineering Materials	By Rangwala	
2.	Engineering Materials	By Deshpande	
3	Engineering Materials	By Ojha	
4	Engineering Materials	By Surendra Singh	
5	Civil Engineering Materials	By T.T.T.I., Madras.	
6	Building Materials	By S.K. Duggal	
7.	Construction Materials	By D.N. Ghose	

BHABHA UNIVERSITY, BHOPAL

Subject –BUILDING CONSTRUCTION

Sub Code – CE3003

UNIT-I

BUILDING COMPONENTS AND MATERIALS

Building components and types of structure - building components & their function. Substructure – foundation, plinth. Superstructure – walls, sill, lintel, doors & windows, floor, roof, parapet, beams, columns. Types of structures – load bearing structures, Framed structures, composite structures.

UNIT-II

CONSTRUCTION OF SUB STRUCTURE

Job layout - necessity and procedures, site clearance, preparing job layout, layout for load bearing structure. Tools and plants used for excavation and earthwork. Foundation - importance and necessity, types of foundation, open foundations, shallow Foundation. Pumping method of dewatering, cofferdams. Bearing capacity of foundation soil.

CONSTRUCTION OF SUPER STRUCTURE

stone masonry - terms used in stone masonry Uncoursed rubble masonry, coursed rubble masonry, point to be observed in construction of stone masonry, mortars for stone masonry, tools and plants used for Stone masonry, col-grout masonry. Brick masonry : common terms used in brick masonry, requirements of good brickwork, bonds in brick masonry. Hollow concrete block masonry, composite masonry, cavity wall- purpose and construction. Doors and windows: doors -components and construction ,Windows -component and construction of fully paneled, partly paneled and glazed, glazed wooden, steel, aluminum windows, sliding windows, louvered window, ventilators, cement grills. Protective treatment

for doors and windows, fixtures and fastenings for doors and window. Selection of doors and windows. Sill, lintel and weather shed - functions, types and construction. Terms used in stair case, characteristics of good staircase, types of staircase - fabricated stair. Scaffolding and shoring: purpose, types of scaffolding, process of erection and dismantling. Merits and demerits of different types of scaffolding.

UNIT-III

BUILDING FINISHES

floors and roofs - floor finishes, brick flooring, flag stone, Shahabad , Kota, marble, granite, Kadappa, ceramic tiles, vitrified, mosaic tiles, Chequerred tiles, glazed tiles, pavement blocks, concrete floors, Tremix floor, skirting and dado. Process of laying- process of laying and construction, finishing and polishing of floors. *Roofing materials* – AC sheets ,GI sheets, plastic sheets, fiber sheets, Mangalore tiles etc. Steel trusses. R.C.C. slab, lean to roof, King post and Queen post trusses (Line Diagram). *Wall finishes*: plastering – necessity of plastering. *Pointing* – necessity and procedure of pointing, Difference between plastering and pointing. *Painting* – necessity, surface preparation, method of application, selecting suitable painting

UNIT-IV

BUILDING MAINTENANCE

Cracks - causes and types of cracks, identification and repair of cracks. Guniting and grouting, use of epoxy and crack fills. *Settlement* - causes and remedial measures plinth protection – necessity and materials used. *Demolition* - necessity, method of demolition, hand demolition, Machine demolition, controlled blasting demolition, precautions during demolition. *Re-baring techniques* - necessity and equipment for re-baring techniques.

UNIT-V

SAFETY AND ENVIRONMENTAL ASPECTS

Safety precautions to be observed during the construction viz. trenching, digging pits for foundation using machineries, masonry works, erection, scaffolding, centering etc. Environmental consideration to be observed during construction of a building e.g. laying out

of drainage line and water supply line, soak pit, septic tank, precautions to be taken during site clearance considering environmental effect. Avoiding unnecessary cutting of bushes and tress etc.

REFERENCES

S. No.	Title	Author	Publisher
1	Building Construction,	S. P. Arora and Bindra,	Dhanpat Rai Publication
2.	Building Construction,	S. C. Rangawala	Charotar Publication
3	Building Construction,	Sushil Kumar,	Standard Publication
4	Building Construction,	B. C. Punmia	Laxmi Publication
5	Building Construction,	S.K. Sharma,	Tata McGraw-Hill
6	Building Construction,	Dr.Janardan Zha,	khanna Publication

Experiment

- Preparing foundation plan and marking on ground layout of load bearing structure by face line method from the given plan of the building.
- Preparing foundations plan and marking on ground layout of framed structure by face line method from the given plan of the building.
- Checking and transferring line and level of plinth, sill, lintel, flooring, slab level of a building and writing report of the process.
- Checking verticality (plumb line) of formwork for column, beam and wall at construction site and writing report of the process.
- Observing and writing report of the process of plastering.
- Observing and writing report of the process of water proofing of terrace or basement.
- Observing the models, specimen of building materials kept in the model room for few building items and writing a report for any five models/materials.
- Visit to a building where slab casting is in progress.
- Use of water level, plum bob, spirit level, Thread, gunia, etc.
- Study and use of various tools used in building construction.

BHABHA UNIVERSITY, BHOPAL

SUBJECT –HYDRAULICS

Sub Code – CE3004

UNIT-I

PROPERTIES OF FLUID

Definition of fluid, Difference in behavior of fluid with respect to solids. Introduction to fluid mechanics and hydraulics, Branches of hydraulics- Hydrostatics and hydrodynamics, Importance of Hydraulics with respect to Irrigation and Environmental engineering. Physical properties of fluid Mass density, Weight density, Specific volume, Specific gravity, Surface tension and capillarity,, Viscosity, Newton's law of viscosity – Dynamic and kinematics viscosity. Ideal and Real liquids

HYDROSTATIC PRESSURE

Free liquid surface, Definition of pressure and its SI unit, Hydrostatic pressure at point, Pascal's law, Variation of pressure in horizontal and vertical direction in static liquid, Pressure diagram. Total hydrostatic pressure and center of pressure, Determination of total pressure & center of pressure on vertical & inclined faces of dams, sluice gates, sides and bottom of water tanks. Numerical Problems.

UNIT-II

MEASUREMENT OF LIQUID PRESSURE IN PIPES

Concept of pressure head and its unit, Intensity of pressure, Variation of pressure with depth of liquid, Types of pressure- atmospheric gauge and absolute pressure. Conversion of pressure head of one liquid in to other, devices for pressure measurements in pipes – Piezometer, U-tube manometer, Bourdon's pressure gauge. Principle of working and limitations. Measurement of pressure difference using differential manometer – U- tube different manometer and inverted U-tube differential manometer. Simple Numerical Problems.

UNIT-III

FUNDAMENTALS OF FLUID FLOW

Concept of flow, Gravity flow and pressure flow. Types of flow – steady and Unsteady,

uniform and non-uniform , Laminar and turbulent. Various combinations of flow with practical examples, Reynolds number and its application, Stream line and equi-potential line. Flow net and its uses. Discharge and its units, Continuity equation for fluid flow. Various forms of energies present in fluid flow-potential, kinetic, & pressure energy. Datum head, pressure head, velocity head and total head, Bernoulli's theorem, its assumptions and limitations. Loss of head and modified Bernoulli's theorem. Application of Bernoulli's theorem. Simple Numerical Problems.

UNIT-IV

FLOW OF LIQUID THROUGH PIPES

Loss of head due to friction, Darcy-Weisbach Equation Friction factor, relative roughness. Common range of friction factor for different types of pipe material. Minor loss of head in pipe flow- loss of head due to sudden Contraction, sudden expansion, gradual contraction & expansion, at entrance and exit of pipes and in various pipe fittings. Pipes in series and parallel, Equivalent pipe – Dupuit's equation. Hydraulic gradient line and Energy gradient line, Siphon pipe. Water hammer in pipes – cause effects and remedial measures, Use of Nomograms for design of water distribution system. Simple Numericals on head loss.

UNIT-V

FLOW THROUGH OPEN CHANNEL

Types of channels- artificial & natural, purposes of artificial channel, Different shapes of artificial channels. Geometrical properties of channel section – wetted area, wetted Perimeter, hydraulics radius. Prismatic channel sections, steady- uniform flow through prismatic channel section. Chezy's equation and Manning's equation for calculation of discharge through an open channel, common range of values of Chezy's constants and Manning's constant of different types of channel surfaces. Most economical channel section, conditions for most economical channel sections. Froude's number and its significance. Critical, sub-critical and supercritical flow in channel, Hydraulic jump its occurrence in field, uses of hydraulic jump

Name of Experiment

Measurement of pressure difference by U-tube differential manometer. Study of bourdon's gauge.

Verification of Bernoulli's theorem.

Reynolds experiment to study types of flow.

Determination of Darcy's friction factor for a given pipe.

Determination of Minor losses in pipes (any two).

Determination of Manning's constant or Chezy's constant for given rectangular channel section.

Demonstration of Hydraulic jump.

Determination of coefficient of discharge for given rectangular or triangular notch.

Determination of coefficient of discharge for a given Venturimeter.

Demonstration and use of Pitot tube and current meter.

Determination of hydraulic coefficients for sharp edge orifice.

Study & use of water meter.

Study of a model of centrifugal and reciprocating pump.

Use of characteristic curves/ charts / catalogs from manufactures for selection of pump for the designed discharge and head (Refer IS: 9694)

REFERENCES

S. No.	Title	Author	Publisher
1	Hydraulics & Fluids Mechanics,	Dr. P.N.Modi & Dr. S.M.Seth,	Standard Book House, Dehli.
2.	Hydraulics & Fluids Mechanics,	S.Ramamrutham,	Dhanpat Rai & Sons, Delhi.
3	A Text Book of Hydraulics,	R.S.Khurmi.	
4	Fluids Mechanics & Hydraulics Machines,		S.Chand & Company Ltd. New, Delhi.
5	A Text Book of Fluids Mechanics Hydraulics Machines,	R.K.Rajput,	S.Chand & Company Ltd. New Delhi.
6	Fluids Mechanics & Hydraulics,	Dr. Jagdish Lal,	Metropolitan Book Co. Private Ltd. New Delhi.
7.	Hydraulics' Laboratory Manual	S.K.Likhi,	T.T.T.I. Chandhigrah.

BHABHA UNIVERSITY, BHOPAL

Subject – BUILDING DRAWING

Sub Code – CE3005

UNIT-I

CONVENTIONS

Conventions as per IS:962-1967 and other practices Types of Lines – Visible line, Centerline, Hidden line, Section line, Dimension line, Extension line, Pointers, Arrow heads or dots. Dimensioning systems. Symbols – Materials used in construction, building components. Reading of available ammonia prints of residential buildings.

UNIT-II

PLANNING OF BUILDING

Principles of planning of Residential and Public building. Space requirements and norms for various units. Residential and Public building. Rules and byelaws of local governing authorities for construction. Drawing lineplans for Residential and Public building.

UNIT-III

BUILDING DRAWING

Development of plan from line plan of a residential building, Elevation, Section, Site plan, Location Plan. Foundation plan, Area statement and other details. Submission Drawing and Working Drawing.

UNIT-IV

DETAILED DRAWING

Drawing of staircase, drawing of steel truss & lean to roof, drawing of layout plan of water supply line with accessories. Layout plan of sanitary line - position of inspection chamber, septic tank, sanitary fittings. Position of wash basin, sink etc.

UNIT-V

PERSPECTIVE DRAWING

Definition, Necessity, Principles of Perspective Drawing, Terms used in perspective drawing, Two point view of a small object like pedestal, step block, small single storied building with flat roof etc.

Name of Experiment

- Drawing various types of lines, lettering and symbols of materials, doors and windows etc. Used in construction on Full Imperial size drawing sheet.
- Drawing the lines plans of following buildings on Full Imperial size graph paper. Residential Building (Min. three rooms)
- Public Building – School building, Primary health center / Hospital building, Bank, Post Office, Hostel building etc.(At least four), Measured Drawing of an existing residential Building (Load bearing/ Framed structure Type) , showing Plan , Elevation, Sections, Construction notes,
- Schedule of openings, Site Plan, Area statement etc
- Submission Drawing of two storied residential building (Framed structure type) showing Plans , Elevation, Sections, Foundation Plan ,construction .
- Working drawing of above drawing sheet preferably one plan, section through stair case to scale 1:50
- Two point perspective view of a building drawn in submission drawing.

REFERENCES BOOK

S. No.	Title	Author	Publisher
1	Text Book of Building Drawing,	Shah, Kale & Patki	
2.	Elements of Building Drawing,	D. M. Mahajan	
3	Planning and Design of Building	Y. S. Sane	Charotar Publishing House Pvt. Ltd.
4	Civil Engineering Drawing	Malik & Meo,	New Asian Publishers New Delhi .
5	Building Drawing,.	V.B.Sikka	

